



RUVARASHE SHALOM MADZIME

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Personal Profile

My MSc thesis studies the formal connection between symbolic machine learning and defeasible reasoning. The central question is: when can a learned rule-based classifier be read as a defeasible theory? I answer this for Exception Closed Conjunctive Rule Sets (ECCRS), a class of classifiers whose predictions are shown to align exactly with Rational Closure, Lexicographic Closure, and System W under a single structural condition called strict global exception closure. This work opens a path toward defeasible machine learning, where learned models are designed not only for predictive accuracy but for principled defeasible semantics. My broader research interests span knowledge representation, interpretable AI, and neuro-symbolic methods.

Education

MSc COMPUTER SCIENCE – University of the Western Cape and CAIR – Western Cape, SA Mar 2025 – Expected Nov 2026

- **Thesis:** Toward a Defeasible Semantics for Symbolic Classifiers
- **Research Area:** Knowledge representation, defeasible reasoning, interpretable machine learning

BSC (HONS) COMPUTER SCIENCE – University of the Western Cape – Western Cape, SA Jan 2024 – Nov 2024

- **Graduated** Cum Laude (85%)
- **Modules:** Artificial Intelligence, Applied Data Science, Internet of Things, Cybersecurity, Big Data Engineering.
- **Research Project:** *Enhanced Electronic Health Records Text Summarisation Using Large Language Models.*

BSC COMPUTER SCIENCE – University of the Western Cape – Western Cape, SA Mar 2021 – Nov 2023

- **Graduated** Cum Laude (84%)
- **Key Modules:** Artificial Intelligence, Mathematics, Statistics.

Publications

- Madzime, R. S., Meyer, T., & Leenen, L. (2025). An Override-Aware Classifier for Transparent AI. *Proceedings of the Southern African Conference for Artificial Intelligence Research (SACAIR) 2025 Conference* (Online Proceedings, Vol. II). ISBN: 978-1-0370-5280-4.
- Madzime, R., & Nyirenda, C. (2024). *Enhanced Electronic Health Record Text Summarization Using Large Language Models*. Proceedings of the Southern African Conference for Artificial Intelligence Research (SACAIR) 2024 Conference. ISBN: 978-0-7961-6069-0.

Awards and Honours

- UWC MasterCard Scholarship (2025 - 2026)
- Top Computer Science Student, University of the Western Cape (2023)
- KnowBe4 Women in Cybersecurity Scholarship (2023)
- Dean's Merit Award, University of the Western Cape (2021, 2022, 2023, 2024)

Research and Professional Experience

RESEARCH ASSISTANT (SARChI NRF) – University of the Western Cape – Western Cape Mar 2026 – Present

- Provide teaching support for undergraduate and postgraduate CS modules under Dr. Leenen's NRF SARChI Chair grant.
- Support supervision of students' Honours projects with research guidance and project feedback.
- Conduct general research assistance in KR as part of the SARChI research programme.

GRADUATE LECTURING ASSISTANT – University of the Western Cape – Western Cape Feb 2025 – Nov 2025

- Supported lecturers with coursework, administrative tasks, and student queries.
- Facilitated tutorial sessions and student consultations for computer science modules.
- Evaluated assignments and examinations.

GRADUATE RESEARCH ASSISTANT – University of the Western Cape – Western Cape Jan 2024 – Nov 2024

- Designed and developed a large language model-based summarization system for Electronic Health Records, enhancing clinician decision-making efficiency.
- Conducted comprehensive dataset analysis using Python and machine learning tools.
- Assisted in securing research funding by identifying gaps in existing literature.

JUNIOR SOFTWARE DEVELOPER INTERN – Apparel Technology– Western Cape Oct 2023 – Mar 2024

- Engineered a Python and Arduino-based system to monitor factory machine uptime and productivity.
- Developed a real-time data visualization framework using Python Dash.
- Managed project tasks effectively with Asana, Teams, and Git.

LEAD STATISTICS TUTOR – University of the Western Cape – Western Cape Feb 2023 – Nov 2023

- Mentored over 50 students, improving pass rates through personalized tutoring.
- Conducted grading and one-on-one sessions to simplify statistical concepts.
- Advocated for course material improvements in collaboration with faculty.

Technical Skills

Programming & Tools: Python, Java, Answer Set Programming (ASP / Clingo), Z3 SMT Solver, LaTeX, Git

Research Areas: Knowledge Representation, Defeasible Reasoning, Interpretable Machine Learning, Nonmonotonic Reasoning, Symbolic AI, NLP, Large Language Models

Supervisors

Associate Professor Louise Leenen – University of the Western Cape & CAIR | lleenen@uwc.ac.za

Professor Thomas Meyer – University of Cape Town & CAIR | tmeyer@cair.org.za